# AI Assisted Coding

## Assignment-16.3

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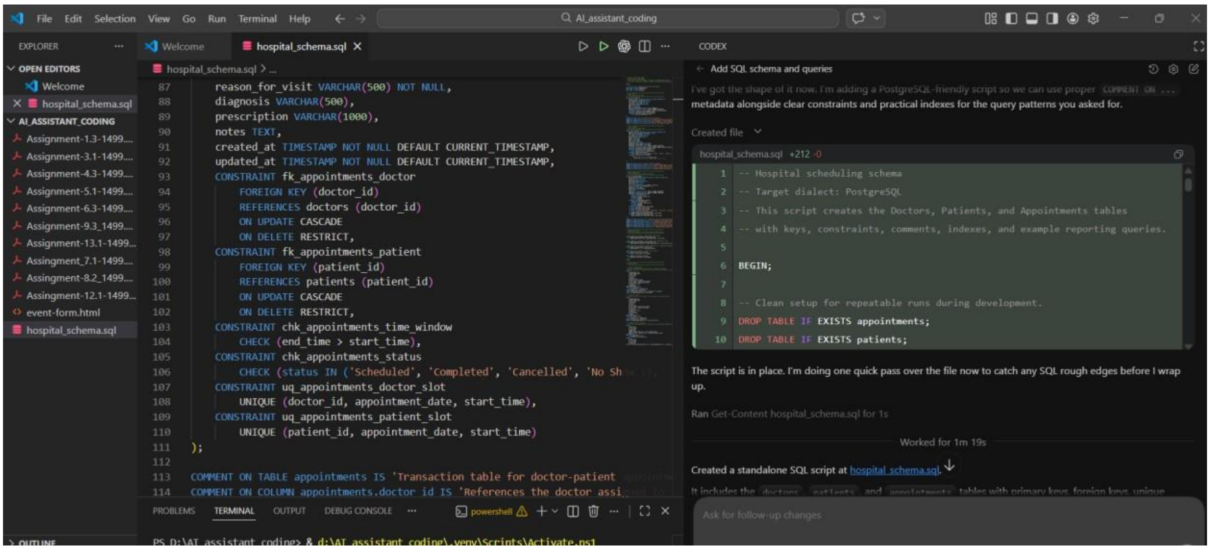
**Batch-08**

## Lab 16 – Database Design and Queries: Schema Design and SQL Generation

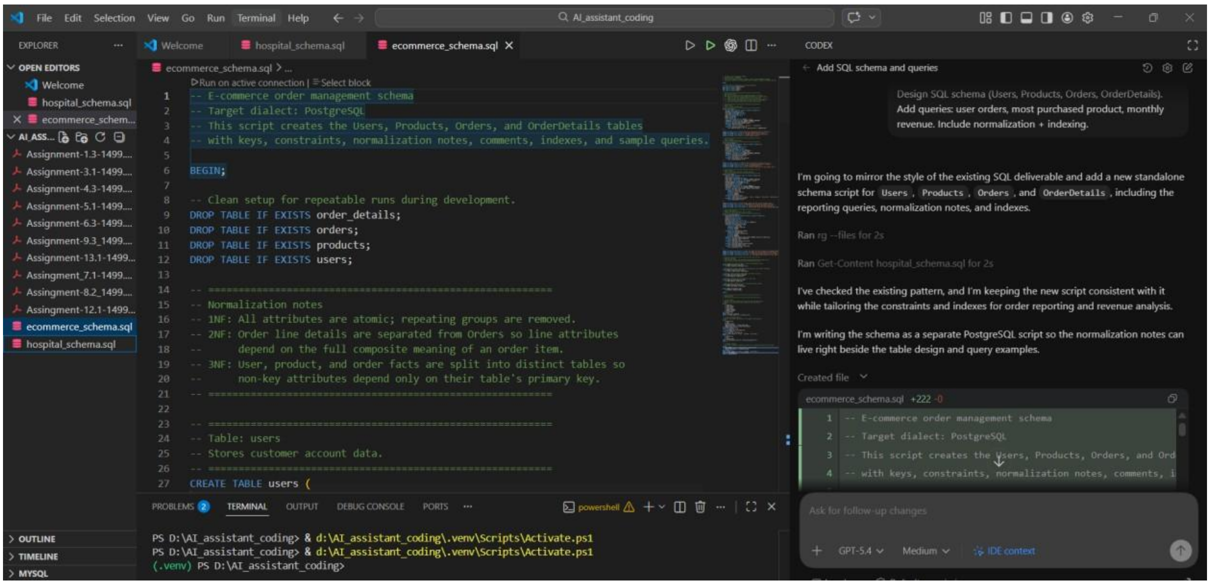
## Task 1 – Hospital Management Database

**Prompt:** "Create SQL schema (Doctors, Patients, Appointments) with keys & constraints. Add queries: doctor appointments, patient history, patient count per doctor. Include comments + indexing."



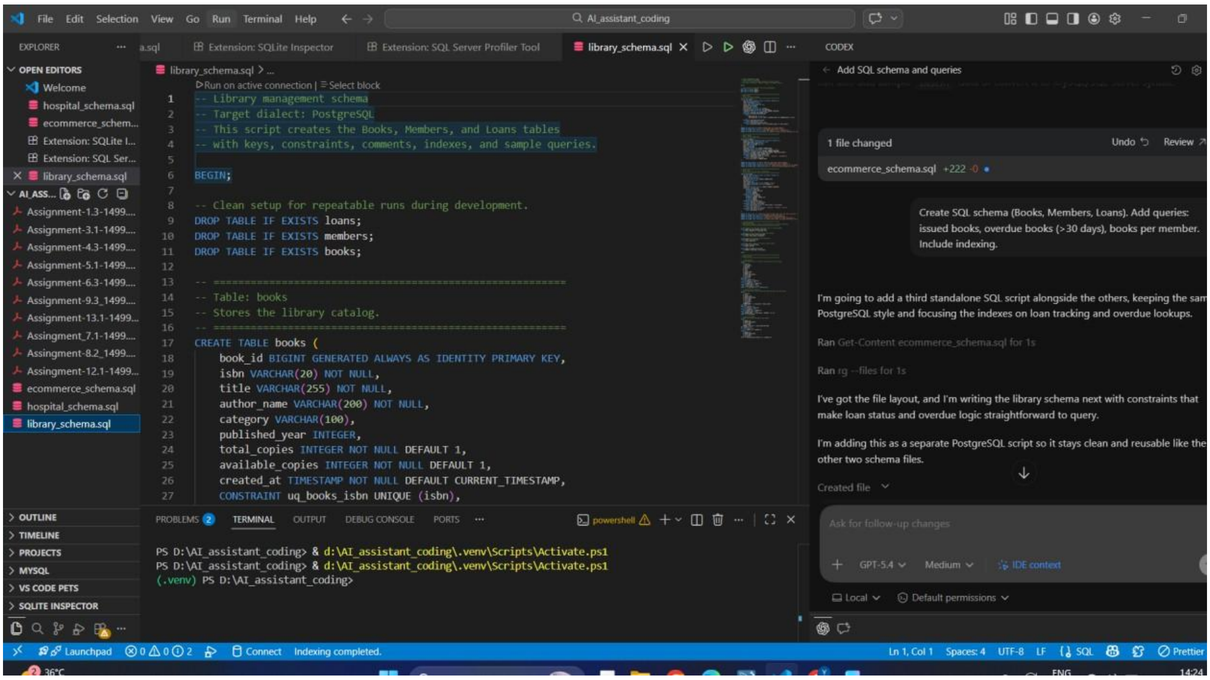

## Task 2 – Real-Time Application: Online Shopping Database

**Prompt:** "Design SQL schema (Users, Products, Orders, OrderDetails). Add queries: user orders, most purchased product, monthly revenue. Include normalization + indexing."



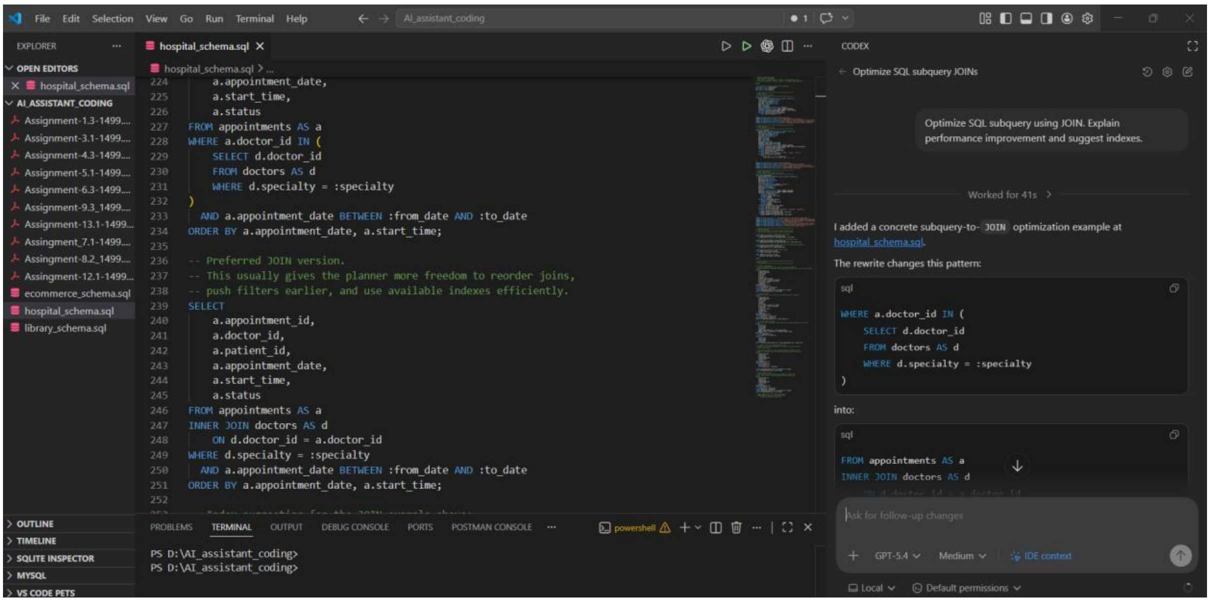

## Task 3 – Library Database

**Prompt:** “Create SQL schema (Books, Members, Loans). Add queries: issued books, overdue books (>30 days), books per member. Include indexing.”




## Task 4: Optimize Queries

**Prompt:** “Optimize SQL subquery using JOIN. Explain performance improvement and suggest indexes.”




## Task 5: University Course Registration

**Prompt:** “Design SQL schema (Students, Courses, Faculty, Registrations). Add queries: students per course, faculty >2 courses, most registered course. Include optimization.”